Math Inquirers

Extension Workbook

with answers

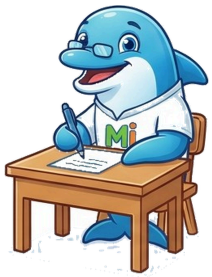
Level 1

Self-study | Ages 6-8



My name _____

Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You do not need a teacher.

For the child

1. Read **Learn it**. Look at the worked example.
2. Do **Start here**. Write in the boxes and on the lines.
3. Try **Think harder**. Stuck? Read the hint. It is upside down!
4. Try **Go further** when you want a big challenge.
5. Do **Check yourself** and count your score.

For the grown-up

- Practice for the same ideas as the Holiday Program, with new numbers and stories.
- About 3 short sessions a week, 20 to 25 minutes each (see the pace plan).
- Let your child try before looking at the answers.
- Answers are at the back, from page 21, with a short reason for each. Read the reason together.
- Check yourself score 2 or less? Repeat Learn it and one Start here page.

The four labels

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are upside down.
- ★ **Go further.** Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

Tips

- Use a pencil. Mistakes help you learn.
- Read each question twice before you write.
- Think harder and Go further are not a race. Come back another day.

Contents

1	**Number patterns** Learn it, practice and check · answers on page 22	page 5
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3	**Picture graphs** Not in this proof edition	–
4	**Repeating and growing patterns** Not in this proof edition	–
5	**Working backwards** Not in this proof edition	–
6	**Money** Not in this proof edition	–
7	**2D shapes** Not in this proof edition	–
8	**Logic and sorting** Not in this proof edition	–
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My pace plan

One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. Tick the box when the week is done.

Week	Unit	Session 1	Session 2	Session 3	Done
1	**Number patterns**	Pages 5-6 Learn it, Start here	Pages 7-8 Start here	Pages 9-11 Think harder, Go further, Check yourself	☐
2	**Number bonds**	Pages 12-13 Learn it, Start here	Pages 14-15 Start here	Pages 16-18 Think harder, Go further, Check yourself	☐
3	**Picture graphs**	Not in this proof edition			☐
4	**Repeating and growing patterns**	Not in this proof edition			☐
5	**Working backwards**	Not in this proof edition			☐
6	**Money**	Not in this proof edition			☐
7	**2D shapes**	Not in this proof edition			☐
8	**Logic and sorting**	Not in this proof edition			☐
9	**Length**	Not in this proof edition			☐
10	**Puzzle day**	Not in this proof edition			☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.			☐

Tips for the grown-up

- Short and often works better than one long session. Stop while your child still wants more.
- Sessions 1 and 2 are the main path. Session 3 has the harder pages: it is fine to do only some of them.
- Missed a week? Just carry on. There is no deadline.

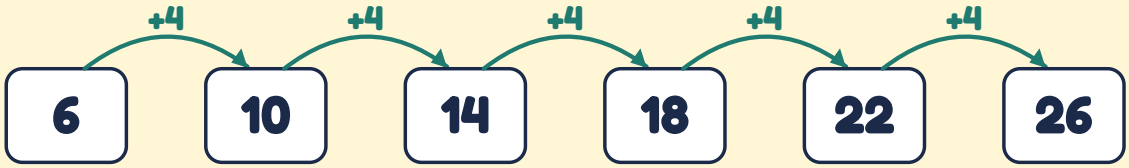
Learn it: Number patterns

My goal. I can find the rule of a number pattern and use it to write the next numbers.

A number pattern follows a rule. The rule tells you how to get from one number to the next. A frog hopping along a number strip makes a pattern.

Worked example

Kumar's frog starts on 6. It hops 4 each time. Write the first 6 numbers.



1. Start on 6. Write 6.
2. Hop 4. 6 and 4 more is 10.
3. Hop 4 again. 10 and 4 more is 14.
4. Keep going: 18, 22, 26. Each number is 4 more than the number before.

Answer: 6, 10, 14, 18, 22, 26. The rule is add 4.

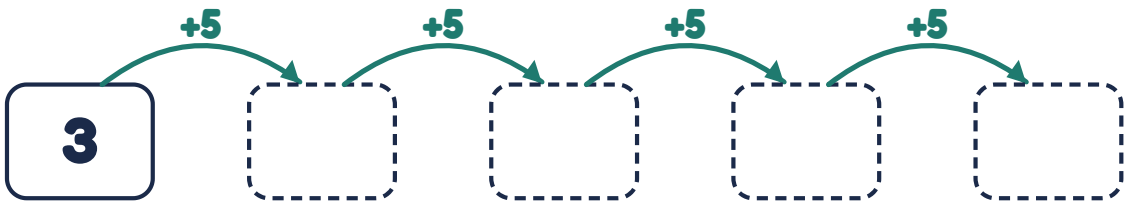
How to find a rule

1. Look at the first 2 numbers. How far is it from one to the other?
2. Check the next 2 numbers. Is the hop the same?
3. Going up? The rule is add. Going down? The rule is take away.
4. Use the rule to write the next numbers.

Your turn

Answer at the back of the book.

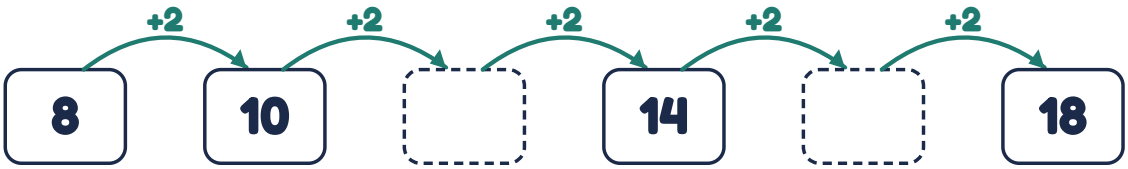
Try **A frog starts on 3 and hops 5 each time. Write the missing numbers in the tiles.**



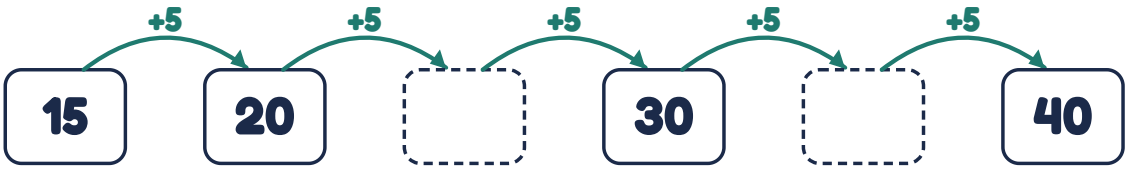
Start here

Do these first. Take your time.

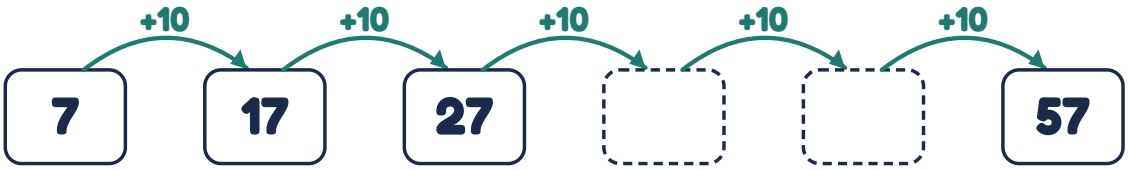
1 Count in 2s. Write the missing numbers in the tiles.



2 Count in 5s. Write the missing numbers in the tiles.



3 Count in 10s. Write the missing numbers in the tiles.



4 Fill in the blanks.

(a) 5, 9, 13, 17, ☐, ☐

(b) The rule is add ☐.

5 Fill in the blanks.

(a) 21, 18, 15, 12, ☐, ☐

(b) The rule is take away ☐.

Start here

Keep going. Check your work.

6 **Circle the rule for this pattern: 7, 11, 15, 19.**

add 3 add 4 add 5

7 **Look at the pattern: 30, 26, 22, 18, ...**

(a) Do the numbers go up or down? Up Down

(b) The rule is take away ☐.

(c) The next number is ☐.

8 **Siti puts 3 stickers in box 1, 6 stickers in box 2 and 9 stickers in box 3. She keeps going.**

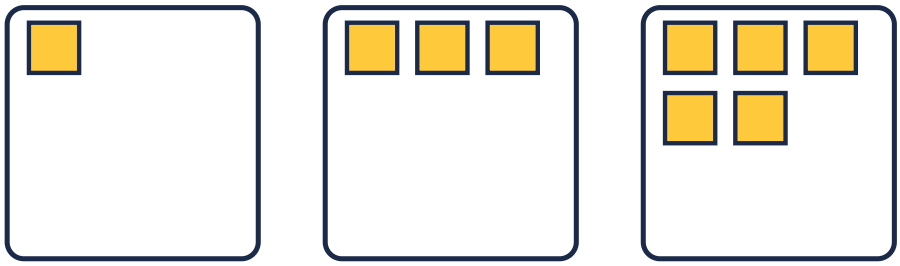
(a) How many stickers go in box 5? ☐ stickers

(b) How many stickers go in box 6? ☐ stickers

Write the numbers: _____

Hint. Each box has 3 more than the one before.

9 **Ben builds steps with blocks.**



Step 1 Step 2 Step 3

(a) How many blocks are in Step 4? ☐ blocks

(b) Each step has ☐ more blocks than the step before.

Start here

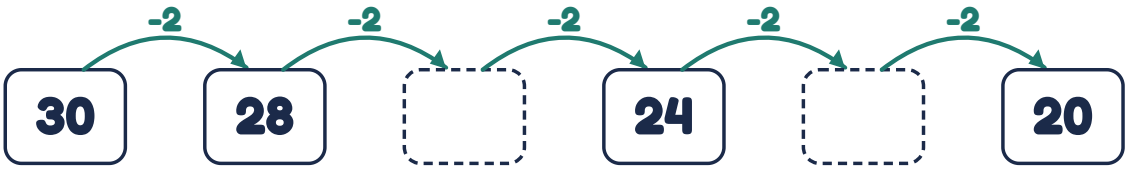
Nearly there. You can do it.

10 One number is wrong. Circle it. Then write the right number.

9, 12, 15, 17, 21, 24

The right number is ☐.

11 Count back in 2s. Write the missing numbers in the tiles.



12 Wei puts 5¢ in his jar on Monday, 10¢ on Tuesday and 15¢ on Wednesday.

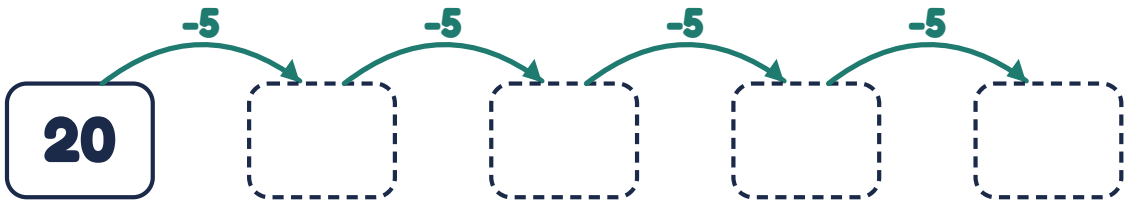
He keeps going. How much does he put in on Friday? ☐ ¢

Write the days and amounts: _____

13 Mei starts on 3 and hops 4 each time. Circle every number she lands on.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

14 Kumar starts on 20 and counts back in 5s. Write the missing numbers in the tiles.



Think harder

Hints are upside down. Try first!

1 **Siti starts on 2 and hops 5 each time.**

(a) Will she land on 37? Yes No

(b) Will she land on 40? Yes No

Write the numbers she lands on: _____

_____

Hint. Write the numbers she lands on, one hop at a time. Stop when you pass 40.

2 **Ali counts in 4s from 4 up to 50. Mei counts in 10s from 10 up to 50.**

(a) Which numbers do they both say? _____

(b) How far apart are the numbers they both say? ☐

Ali: _____

Mei: _____

Hint. Write Ali's numbers on one line and Mei's numbers on the other. Look for matches.

3 **A pattern adds the same number each time. Its 2nd number is 10. Its 5th number is 25.**

(a) How many hops from the 2nd number to the 5th number? ☐

(b) The rule is add ☐.

(c) The 1st number is ☐.

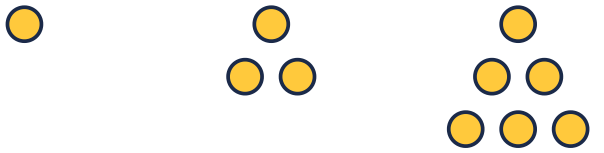
Hint. From 10 to 25 is 15. Share 15 equally between the hops.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Triangle numbers

1 Look at the triangles made of dots. Triangle 1 has 1 dot, Triangle 2 has 3 dots and Triangle 3 has 6 dots. The pattern keeps going.



Triangle 1 Triangle 2 Triangle 3

(a) Triangle 4 has ☐ dots.

(b) Triangle 5 has ☐ dots.

(c) Triangle 7 has ☐ dots.

Hint. How many more dots does each triangle have than the one before? Is it always the same?

2 Mei starts on 1 and adds 3 each time. Ali starts on 2 and adds 5 each time. Both stop at 40. Write the numbers that Mei and Ali both say.

Mei: _____

Ali: _____

Both: _____

Hint. Write each pattern on its own line. Then look for the same number in both lines.

3 A pattern starts on 2 and adds the same whole number each time. The pattern says 20. Which numbers can it add? Find them all.

I tried: _____

_____

Hint. 20 is 18 more than 2. Which equal hops make exactly 18?

✔ Check yourself

No hints. Try each one on your own.

1 14, 18, 22, 26, ☐, ☐

Working: _____

2 40, 35, 30, 25, ☐, ☐. The rule is take away ☐.

Working: _____

3 Is 30 in this pattern: 6, 10, 14, 18, ...? Yes No

Working: _____

4 One number is wrong: 11, 16, 21, 27, 31, 36. The right number is ☐.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

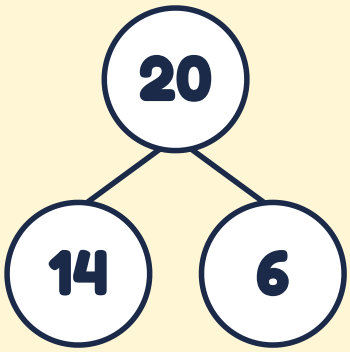
Learn it: Number bonds

My goal. I can find the missing partner in a bond to 10 and to 20. I can tell when the 2 sides are equal.

A number bond is 2 numbers that make a whole. 6 and 4 make 10. So 6 + 4 = 10 and 10 = 6 + 4. Bonds to 10 help us find bonds to 20.

Worked example

Aisha has 20 beads. 14 beads are blue and the rest are white. How many beads are white?



20 = 14 + 6

1. The whole is 20. One part is 14. We want the other part.
2. 14 is 10 and 4 more. We need 4 and some more to make 10: 4 + 6 = 10.
3. So 14 + 6 = 20. The other part is 6.
4. Check: 14 + 6 = 20.

Answer: 6 beads are white. 20 = 14 + 6.

How to find the missing partner

1. Find the whole (the greatest number).
2. Look at the ones digit of the part you know. What makes 10 with it?
3. Add the partners. Do they make the whole?

Your turn

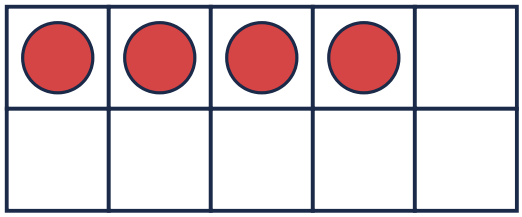
Answer at the back of the book.

Try **Fill in the blank. 20 = 17 + ☐**

Start here

Do these first. Take your time.

1 **Look at the ten-frame.**



(a) How many more counters fill the frame? ☐ counters

(b) 10 = 4 + ☐

2 **Fill in the missing numbers.**

(a) 10 = 8 + ☐

(b) 10 = ☐ + 3

(c) 10 = 5 + ☐

(d) 10 = 1 + ☐

3 **Kumar has 10 marbles. 6 marbles are in his pocket. The rest are in a bag.**

How many marbles are in the bag? ☐ **marbles**

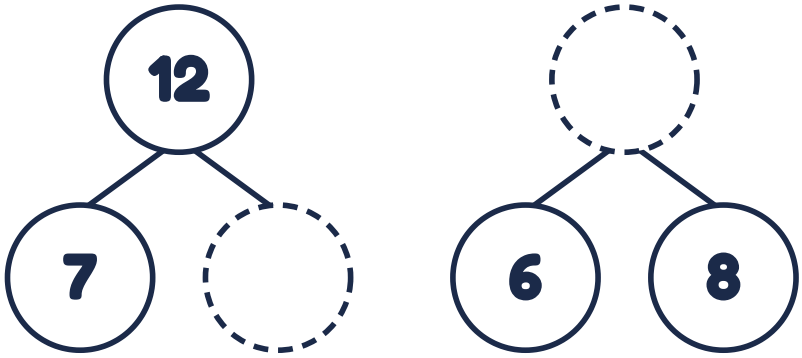
4 **Write the partner that makes 10.**

1 and ☐ **3 and** ☐ **6 and** ☐ **8 and** ☐

Start here

Keep going. Check your work.

5 Write the missing number in each bond.



6 Fill in the missing numbers.

(a) 20 = 15 + ☐

(b) 20 = ☐ + 14

(c) 20 = 18 + ☐

(d) 20 = 11 + ☐

Hint. Look at the ones digit. What makes 10 with it?

7 7 + 3 = 10. Use it to fill in the blanks.

17 + ☐ = 20 27 + ☐ = 30 47 + ☐ = 50

8 Mei has a shelf for 20 books. There are 16 books on the shelf. How many more books fit? ☐ books

Start here

Nearly there. You can do it.

9 **Fill in the missing number. Both sides must be equal.**

(a) 6 + 5 = 7 + ☐

(b) 8 + 2 = 5 + ☐

(c) 9 + 4 = ☐ + 5

10 **Look at the pairs that make 9.**

(a) 1 + 8 = 9 2 + ☐ = 9 3 + ☐ = 9 4 + ☐ = 9

(b) The first numbers go up by 1. The second numbers go up down by 1.

11 **Circle True or False.**

(a) 4 + 6 = 10 True False

(b) 8 + 3 = 10 True False

(c) 13 + 7 = 20 True False

(d) 15 + 6 = 20 True False

12 **There are 20 children in a club. 7 children are girls. The rest are boys. How many boys are in the club?** ☐ **boys**

13 **Kumar has 8 stickers.**

(a) He wants 10 stickers. How many more does he need? ☐ stickers

(b) He wants 20 stickers. How many more does he need? ☐ stickers

Think harder

Hints are upside down. Try first!

1 **Ravi puts 14 marbles into 2 bags. The first bag has fewer marbles than the second bag. Each bag has at least 1 marble. Write every way in the table.**

How many ways are there? ☐

First bag	Second bag

Hint. Try 1 marble in the first bag, then 2. Stop when the first bag is not fewer.

2 **Each shape stands for a number. The same shape is always the same number.**

(a) ★ + ★ + 5 = 17. ★ = ☐

(b) ● + ● + ● = 21. ● = ☐

Hint. Take 5 away from 17 first. Then share what is left between the 2 stars.

3 **Three different numbers from 1 to 6 add up to 9. Find every set of three.**

☐ + ☐ + ☐ = 9

☐ + ☐ + ☐ = 9

☐ + ☐ + ☐ = 9

Hint. Write the numbers from smallest to greatest. Start with 1 as the smallest number.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Making two amounts equal

1 **Ali has 16 stickers. Mei has 8 stickers. Ali gives some of his stickers to Mei. Now each child has an equal number of stickers.**

(a) How many stickers does Ali give Mei? ☐ stickers

(b) How many stickers does each child have now? ☐ stickers

Hint. Put all the stickers together. Share them equally between the 2 children.

2 **Mei puts 17 sweets in 2 jars. Jar A has 5 more sweets than Jar B.**

(a) Jar A has ☐ sweets.

(b) Jar B has ☐ sweets.

Hint. Take away the 5 extra sweets. Share what is left between the 2 jars.

3 **I am a number. When I add 7, the answer ends in 0 and is between 15 and 45. What can I be? Find them all.**

The answer can be: _____

So I can be: _____

Hint. First write the numbers between 15 and 45 that end in 0.

4 **Pick 2 different numbers from 1, 2, 3, 4, 5 and add them.**

(a) Write every total you can make. Begin with the smallest.

_____

(b) Which totals can you make in only one way? _____

My pairs: _____

_____

Hint. List the pairs in order: 1 + 2, 1 + 3, 1 + 4, 1 + 5, then 2 + 3 …

✔ Check yourself

No hints. Try each one on your own.

1 $10 = 2 + \square$

Working: _____

2 $20 = 6 + \square$

Working: _____

3 $7 + 5 = 9 + \square$

Working: _____

4 There are 20 flowers. 11 are red. The rest are yellow. How many flowers are yellow? ☐ flowers

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number patterns	☐	☐	☐	☐	/ 4	
2	Number bonds	☐	☐	☐	☐	/ 4	
3	Picture graphs	☐	☐	☐	☐	/ 4	
4	Repeating and growing patterns	☐	☐	☐	☐	/ 4	
5	Working backwards	☐	☐	☐	☐	/ 4	
6	Money	☐	☐	☐	☐	/ 4	
7	2D shapes	☐	☐	☐	☐	/ 4	
8	Logic and sorting	☐	☐	☐	☐	/ 4	
9	Length	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 1.

Date

Signed by a grown-up

Answers

For grown-ups. Please let your child try each question first, then read the short reason together.

Printed copy: this section is at the end of the book. If you want to keep the answers apart while your child works, take these pages out along the dashed line in the margin.

Unit 1 · Hops and rules

Your turn 8, 13, 18, 23
Why: Add 5 each time: 3 + 5 = 8, 8 + 5 = 13, 13 + 5 = 18, 18 + 5 = 23.

START HERE

1 12, 16
Why: Each hop is 2. 10 and 2 more is 12. 14 and 2 more is 16.

2 25, 35
Why: Each hop is 5. 20 and 5 more is 25. 30 and 5 more is 35.

3 37, 47
Why: Each hop is 10. The tens digit goes up by 1 and the ones digit stays 7.

4 **(a)** 21, 25 **(b)** add 4
Why: From 5 to 9 is 4 more, and from 9 to 13 is 4 more. So the rule is add 4: 17 + 4 = 21, 21 + 4 = 25.

5 **(a)** 9, 6 **(b)** take away 3
Why: The numbers go down by 3 each time: 12 - 3 = 9 and 9 - 3 = 6.

6 add 4
Why: 7 to 11 is 4 more. 11 to 15 is 4 more. 15 to 19 is 4 more.

7 **(a)** Down **(b)** take away 4 **(c)** 14
Why: 30 is more than 26, so the numbers go down. 30 - 26 = 4, so take away 4. 18 - 4 = 14.

8 **(a)** 15 stickers **(b)** 18 stickers
Why: 3, 6, 9, 12, 15, 18. The 4th box has 12, the 5th has 15 and the 6th has 18.

9 **(a)** 7 blocks **(b)** 2
Why: Step 1 has 1 block, Step 2 has 3 and Step 3 has 5. Each step adds 2 blocks, so Step 4 has 5 + 2 = 7.

10 17 is wrong. The right number is 18.
Why: The pattern adds 3 each time: 9, 12, 15, 18, 21, 24. So 17 should be 18.

11 26, 22
Why: Counting back means taking away 2 each time: 28 - 2 = 26 and 24 - 2 = 22.

12 25¢
Why: Each day he puts in 5¢ more: Thursday 20¢, Friday 25¢.

13 3, 7, 11, 15, 19, 23, 27, 31
Why: Start on 3 and count on 4 each time: 3, 7, 11, 15, 19, 23, 27, 31.

14 15, 10, 5, 0
Why: Counting back in 5s means taking away 5 each time: 20 - 5 = 15, 15 - 5 = 10, 10 - 5 = 5, 5 - 5 = 0.

THINK HARDER

1 **(a)** Yes **(b)** No
Why: 2, 7, 12, 17, 22, 27, 32, 37, 42. She lands on 37. She jumps from 37 over 40 to 42.

2 **(a)** 20 and 40 **(b)** 20
Why: Ali says 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48. Mei says 10, 20, 30, 40, 50. They both say 20 and 40, and 40 - 20 = 20.

3 **(a)** 3 hops **(b)** add 5 **(c)** 5
Why: From the 2nd to the 5th number is 3 hops (2nd to 3rd, 3rd to 4th, 4th to 5th). They make 25 - 10 = 15, so each hop is 15 / 3 = 5. Go back one hop from 10: the 1st number is 5.

GO FURTHER

1 **(a)** 10 dots **(b)** 15 dots **(c)** 28 dots
Why: Each triangle has one more row than the one before, and the new row has one more dot. So the extra dots are 2, 3, 4, 5, 6, 7: 1, 3, 6, 10, 15, 21, 28.

2 7, 22 and 37
Why: Mei says 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, 37, 40. Ali says 2, 7, 12, 17, 22, 27, 32, 37. The same numbers are 7, 22 and 37, and each is 15 more than the one before.

3 1, 2, 3, 6, 9, 18
Why: You need equal hops that make 18 from 2 to 20. Hops of 1 (18 of them), 2 (9), 3 (6), 6 (3), 9 (2) and 18 (1) all work.

CHECK YOURSELF

1 30, 34
Why: The rule is add 4: 26 + 4 = 30 and 30 + 4 = 34.

2 20, 15; take away 5
Why: The numbers go down by 5 each time: 25 - 5 = 20 and 20 - 5 = 15.

3 Yes
Why: Add 4 each time: 6, 10, 14, 18, 22, 26, 30. It lands on 30.

4 27 is wrong. The right number is 26.
Why: The pattern adds 5: 11, 16, 21, 26, 31, 36.

Unit 2 · Make 10 and 20

Your turn 3
Why: 7 and 3 make 10, so 17 and 3 make 20. Check: 17 + 3 = 20.

START HERE

1 **(a)** 6 counters **(b)** 10 = 4 + 6
Why: The frame has 10 spaces. 4 are full, so 6 are empty. 4 + 6 = 10.

2 **(a)** 2 **(b)** 7 **(c)** 5 **(d)** 9
Why: Think of the partner that makes 10: 8 and 2, 7 and 3, 5 and 5, 1 and 9.

3 4 marbles
Why: 6 and 4 make 10, so 4 marbles are in the bag.

4 9, 7, 4, 2
Why: 1 + 9 = 10, 3 + 7 = 10, 6 + 4 = 10, 8 + 2 = 10.

5 5 and 14
Why: First bond: 12 = 7 + 5. Second bond: the two parts 6 and 8 make 14.

6 **(a)** 5 **(b)** 6 **(c)** 2 **(d)** 9
Why: 5 + 5 = 10, 4 + 6 = 10, 8 + 2 = 10, 1 + 9 = 10. So the partners to 20 are 5, 6, 2 and 9.

7 3, 3, 3
Why: The ones digit is 7 each time. 7 + 3 = 10, so each number needs 3 more to reach the next ten.

8 4 books
Why: 16 + 4 = 20, because 6 + 4 = 10.

9 **(a)** 4 **(b)** 5 **(c)** 8
Why: The sign = means both sides have the same value. 6 + 5 = 11, and 7 + 4 = 11. 8 + 2 = 10, and 5 + 5 = 10. 9 + 4 = 13, and 8 + 5 = 13.

10 **(a)** 7, 6, 5 **(b)** down
Why: 2 + 7, 3 + 6 and 4 + 5 all make 9. When one number goes up by 1, the other goes down by 1, so the total stays 9.

11 **(a)** True **(b)** False **(c)** True **(d)** False
Why: 8 + 3 = 11, not 10. 15 + 6 = 21, not 20. The other two are correct bonds.

12 13 boys
Why: 7 + 3 = 10, so 7 + 13 = 20. There are 13 boys.

13 **(a)** 2 stickers **(b)** 12 stickers
Why: 8 + 2 = 10. From 10 to 20 is 10 more, so 8 + 12 = 20.

THINK HARDER

1 6 ways
Why: The ways are 1 and 13, 2 and 12, 3 and 11, 4 and 10, 5 and 9, 6 and 8. 7 and 7 is not allowed because the first bag must have fewer.

2 **(a)** ★ = 6 **(b)** ● = 7
Why: (a) 17 - 5 = 12, and 6 + 6 = 12, so ★ = 6. (b) Three equal numbers make 21: 7 + 7 + 7 = 21, so ● = 7.

3 1 + 2 + 6, 1 + 3 + 5 and 2 + 3 + 4
Why: With 1 as the smallest: 1 + 2 + 6 and 1 + 3 + 5. With 2 as the smallest: 2 + 3 + 4. The smallest number cannot be 3 or more, because 3 + 4 + 5 = 12.

GO FURTHER

1 **(a)** 4 stickers **(b)** 12 stickers
Why: 16 + 8 = 24 stickers. Shared equally, each child has 12. Ali goes from 16 to 12, so he gives 4.

2 **(a)** 11 sweets **(b)** 6 sweets
Why: 17 - 5 = 12. Shared equally, each jar gets 6. Jar A also has the 5 extra: 6 + 5 = 11. Check: 11 + 6 = 17.

3 13, 23 and 33
Why: The numbers that end in 0 between 15 and 45 are 20, 30 and 40. Take away 7 from each: 13, 23 and 33.

4 **(a)** 3, 4, 5, 6, 7, 8, 9 **(b)** 3, 4, 8 and 9
Why: The pairs are 1+2=3, 1+3=4, 1+4=5, 1+5=6, 2+3=5, 2+4=6, 2+5=7, 3+4=7, 3+5=8, 4+5=9. The totals are 3 to 9. 5, 6 and 7 can each be made two ways. 3, 4, 8 and 9 only one way.

CHECK YOURSELF

1 8
Why: 2 and 8 make 10.

2 14
Why: 6 and 4 make 10, so 6 and 14 make 20.

3 3
Why: 7 + 5 = 12. 9 and 3 make 12.

4 9 flowers
Why: 11 + 9 = 20, because 1 + 9 = 10.